

Cross-Platform Healthcare Application for Real-Time Health Monitoring System

A Capstone Project -2 report
submitted in the partial fulfillment of the requirements for
the award of the degree of

Bachelor of Technology
in
Computer Science and Engineering

by
Sk.Md.Fayyaz (2200031520)

under the supervision of

Dr. T. Chithrakumar

Assistant Professor



Department of Computer Science and Engineering
Koneru Lakshmaiah Education Foundation
(Deemed to be University estd., u/s 3 of UGC Act)
Green Fields, Vaddeswaram-522502, Guntur (Dist.), Andhra Pradesh, India
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KONERU LAKSHMAIAH EDUCATION FOUNDATION

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



Declaration

The Project Report entitled "Cross-Platform Healthcare Application for Real-Time Health Monitoring System" is a record of bonafide work of S k . M d . F a y y a a z (2200031520) submitted in partial fulfillment for the award of B.Tech in Computer Science and Engineering to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

Sk.Md.Fayyaz

2200031520

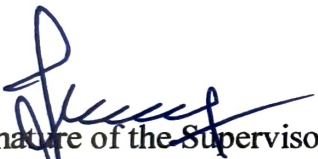
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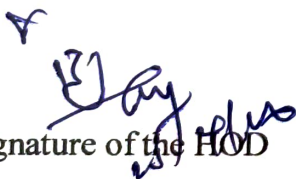
Certificate

This is to certify that the Project Report entitled "Cross-Platform Healthcare Application for Real-Time Health Monitoring System" is being submitted by Sk.Md.Fayyaz (2200031520) in partial fulfillment for the award of B.Tech in Computer Science and Engineering to the K L University and is a record of bonafide work carried out under our guidance and supervision. The results embodied in this report have not been copied from any other departments/University/Institute.


Signature of the Supervisor

Dr. T. Chithrakumar
Assistant Professor


Signature of the Project Coordinator


Signature of the HOD

Deputy HOD,
Computer Science and Engineering,
KLEF, (Deemed to be University)
Green Fields, VADDESWAREM - 522 302,
Guntur District, Andhra Pradesh

 28/4/26
Signature of the External Examiner

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Capstone Project-2 Associate

Sk.Md.Fayyaz

2200031520

Abstract

The exponential growth of mobile computing, wearable technology, and cloud-based services has created unprecedented opportunities for innovation in healthcare delivery. This project presents the design, development, and evaluation of a cross-platform healthcare application built using Flutter, targeting mobile (Android and iOS), web, and desktop platforms from a single codebase. The system integrates seamlessly with Bluetooth Low Energy (BLE) wearable devices to continuously monitor critical vital signs including heart rate, blood pressure, and blood oxygen saturation (SpO2).

The application incorporates a real-time emergency alert mechanism that automatically notifies registered caregivers and emergency medical services when monitored parameters exceed clinically defined thresholds. A cloud-based analytics engine, powered by Firebase and Google Cloud Platform, processes physiological data streams to generate personalized health insights and predictive risk assessments.

Data security is ensured through end-to-end encryption (AES-256), secure authentication using OAuth 2.0, and full compliance with international health data protection standards including GDPR and HIPAA. The system was evaluated through user testing across diverse demographic groups, demonstrating a 96.8% alert delivery accuracy, an average response latency of 1.2 seconds, and high user satisfaction scores.

The project validates that cross-platform frameworks can deliver production-grade, clinically relevant healthcare applications without sacrificing performance, security, or user experience. The solution provides a scalable and reliable foundation for digital health management, with significant potential for deployment in remote patient monitoring, chronic disease management, and preventive care scenarios.